Example 4.57 Out of 7 consonants and 4 vowels, how many strings of 3 consonants and 2 vowels can be formed?

Solution:
Number of ways of selecting (3 consonants out of 7) and (2 vowels out of 4) is

$$^7C_3 \times^4 C_2$$

Each string contains 5 letters. Number of ways of arranging 5 letters among themselves is $5! = 120$
Hence required number of ways is,

$$^7C_3 \times^4 C_2 \times 5! = 35 \times 6 \times 120 = 25200$$

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Example 4.58 Find the number of strings of 5 letters that can be formed with the letters of the word PROPOSITION.

Solution:
There are 11 letters in the word, with respect to number of repetitions of letters there are 4 distinct letters (R, S, T, N), 2 sets of two alike letters (PP,II), 1 set of three alike letters (OOO). The following table will illustrate the combination of these sets and the number of words

S.No	Letter Options	Selections	Arrangements
1	5 distinct (R, S, T, N, P, I, O)	$^7C_5$	$^7C_5 \times 5! = 2520$
2	1 set of 3 alike (OOO), 1 set of 2 alike (PP, II)	$^1C_1 \times^2 C_1$	$^1C_1 \times^2 C_1 \times \frac{5!}{3!\times 2!} = 20$
3	1 set of 3 alike (OOO), 2 distinct (R, S, T, N, P, I)	$^1C_1 \times^6 C_2$	$^1C_1 \times^6 C_2 \times \frac{5!}{3!} = 300$
4	2 sets of 2 alike (PP, II, OO), 1 distinct $(R, S, T, N$ and remaining one in 2 alike)	$^3C_2 \times^5 C_1$	$^3C_2 \times^5 C_1 \times \frac{5!}{2!\times 2!} = 450$
5	1 set of 2 alike (PP, II, OO), 3 distinct $(R, S, T, N$ and remaining two in 2 alike)	$^3C_1 \times^6 C_3$	$^3C_1 \times^6 C_3 \times \frac{5!}{2!} = 3600$

Hence, the total number of strings are $2520 + 20 + 300 + 450 + 3600 = 6890$.

Example 4.59 If a set of m parallel lines intersect another set of n parallel lines (not parallel to the lines in the first set), then find the number of parallelograms formed in this lattice structure.

Solution:
Whenever we select 2 lines from the first set of m lines and 2 lines from the second set of n lines, one parallelogram is formed. Thus the number of parallelograms formed is $^mC_2 \times^n C_2$.